

Hydraulic power system — General description

General

The hydraulic power system supplies pressurized fluid to the flight control actuators, the landing gear retraction jacks, the wheel brakes and the nose wheel steering unit. The system is arranged as two independent circuits, referred to throughout this publication as the green circuit and the yellow circuit, so that the loss of one circuit does not remove hydraulic power from any flight-critical function.

Each circuit has its own reservoir, engine-driven pump, filter assembly and accumulator. The circuits share no fluid path in normal operation. A power transfer unit couples the two circuits mechanically, without transferring fluid, so that a circuit whose pump has failed can still be pressurized from the opposite side.

Do not disconnect any hydraulic line before the circuit has been depressurized and the accumulator has been discharged. Residual pressure in an accumulator can be sufficient to cause injury.

Nominal system pressure is 3000 psi. The relief valve in each circuit opens at 3500 psi and reseats at 3200 psi. Reservoir pressurization is taken from engine bleed air through a dedicated regulator; on the ground, with the engines shut down, the reservoirs are pressurized by the hand pump on the ground service panel.

All pressures quoted in this data module are gauge pressures measured at the system pressure transducer unless another location is stated.

Component description

The main components of each circuit are listed below. Part numbers are given in the illustrated parts data module for the system.

Reservoir

The reservoir is a welded aluminium cylinder of 12 litre usable capacity with a bootstrap piston. The piston is driven by system pressure so that the return side of the circuit is held above the cavitation threshold of the engine-driven pump at all attitudes and at all altitudes within the flight envelope.

A sight glass on the forward face of the reservoir shows the fluid level against three marks: REFILL, FULL and MAX. The level must be read with the aircraft on the ground, the circuit depressurized and the landing gear extended.

Engine-driven pump

The engine-driven pump is a variable displacement axial piston unit mounted on the accessory gearbox. Displacement is controlled by a compensator that maintains delivery pressure at the nominal value across the whole flow range. At zero demand the pump destrokes and the delivery flow falls to the case drain flow only.

Running the pump dry for more than fifteen seconds will damage the piston shoes. Prime the circuit before the first engine start after any component change.

Accumulator

Each circuit has a piston accumulator precharged with dry nitrogen. The accumulator damps pressure transients caused by rapid actuator movement and provides a reserve of stored energy for brake application after the loss of both pumps.

Green circuit accumulator, 1.5 litre, precharged to 1800 psi.

Yellow circuit accumulator, 1.0 litre, precharged to 1800 psi.

Brake accumulator, 0.5 litre, precharged to 1200 psi.

Leading particulars

The table that follows gives the values used when the system is set up after installation. Values marked with an asterisk are checked at every scheduled inspection.

System leading particulars

Parameter Value Tolerance

System pressure * 3000 psi +/- 50 psi

Relief valve cracking pressure 3500 psi +/- 75 psi

Reservoir capacity 12 L +/- 0.2 L

Accumulator precharge * 1800 psi +/- 25 psi

Case drain flow at idle 1.2 L/min max

Fluid conforming to the specification quoted in the consumables data module must be used throughout. Fluids of different specifications must not be mixed, and a circuit that has been contaminated must be drained, flushed and refilled before it is returned to service.